SOC Lab Project – Implementation & Investigation

1. Project Introduction

This project demonstrates the implementation of a practical Security Operations Center (SOC) environment using Splunk Enterprise, Prometheus and Grafana.

The project focuses on centralized log collection, security event analysis, controlled attack simulation, detection engineering, alert generation, infrastructure monitoring and incident response.

A controlled SSH brute-force scenario was performed to demonstrate the complete SOC investigation workflow, from generating and detecting suspicious authentication activity to analyzing security events and documenting the incident response findings.

2. Project Architecture

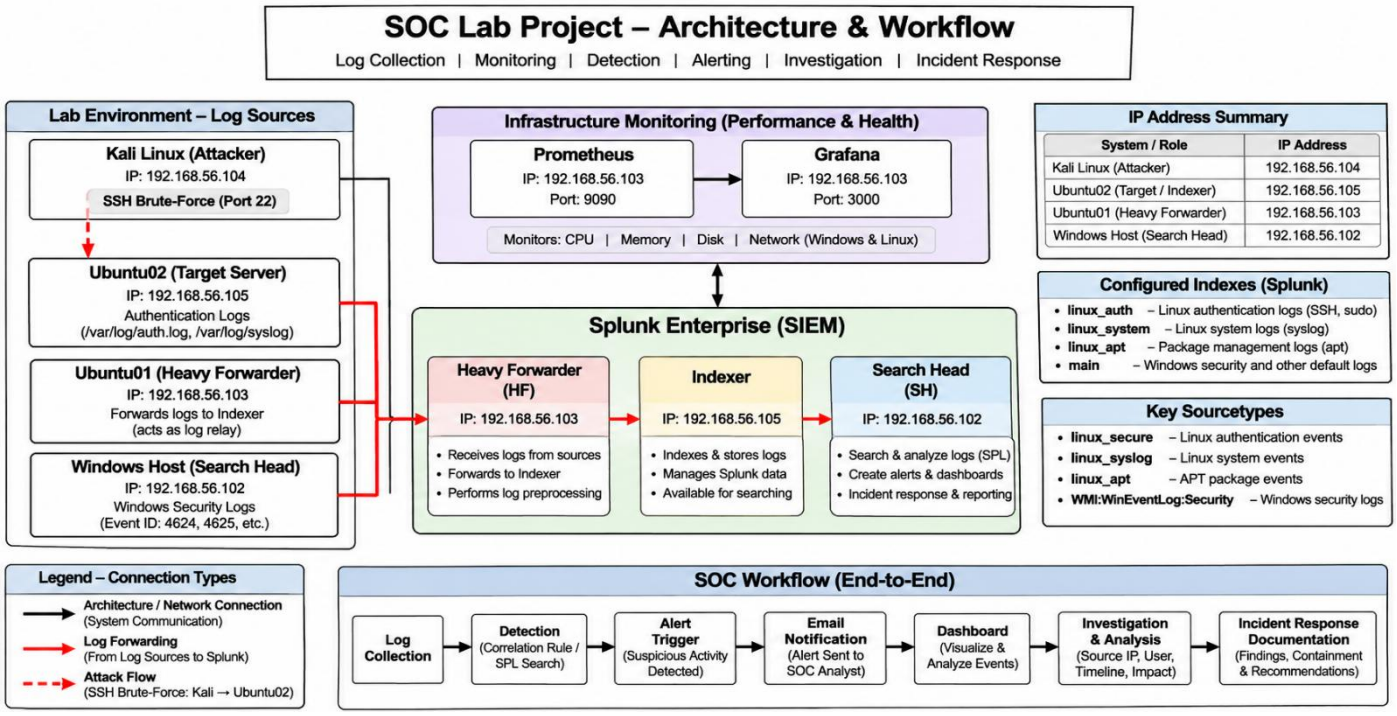


Figure: SOC Lab Project – Architecture and Workflow.

*For the initial lab setup, network configuration, Splunk installation and component configuration, refer to the separate **Phase 1 – Splunk Lab Installation & Configuration Documentation**.*

The architecture diagram provides an overview of the complete SOC lab environment, including its infrastructure, network communication, log collection, security monitoring and incident response workflow. Kali Linux acts as the attack simulation machine, generating controlled SSH brute-force attempts against Ubuntu02, which serves as the target server. The authentication logs from Ubuntu02 are forwarded to Ubuntu01, configured as the Splunk Heavy Forwarder, and then transferred to the Splunk Indexer for storage and processing. The Windows host operates as the Splunk Search Head, providing the interface for searching, analyzing and visualizing the collected security events.

The diagram also illustrates the integration of Prometheus and Grafana for monitoring infrastructure performance, including CPU, memory, disk and network metrics. The IP address summary, configured Splunk indexes and key sourcetypes provide an overview of the lab's network and log organization.

3. SSH Brute-Force Attack Simulation

A controlled SSH brute-force simulation was performed from Kali Linux against the Ubuntu02 target machine using the Hydra tool and a custom Bash script.

The script automated multiple SSH authentication attempts against the soc_test account using a predefined password list. This simulation was conducted within the isolated SOC lab environment to generate failed authentication events for security monitoring and detection.

Attack Source: Kali Linux – 192.168.56.104
Target System: Ubuntu02 – 192.168.56.105
Target Account: soc_test
Tool Used: Hydra

Purpose of the Test:
The objective was to generate controlled SSH authentication failures and verify whether the target system recorded these events and whether they could subsequently be collected, monitored and analyzed through the Splunk SIEM environment.

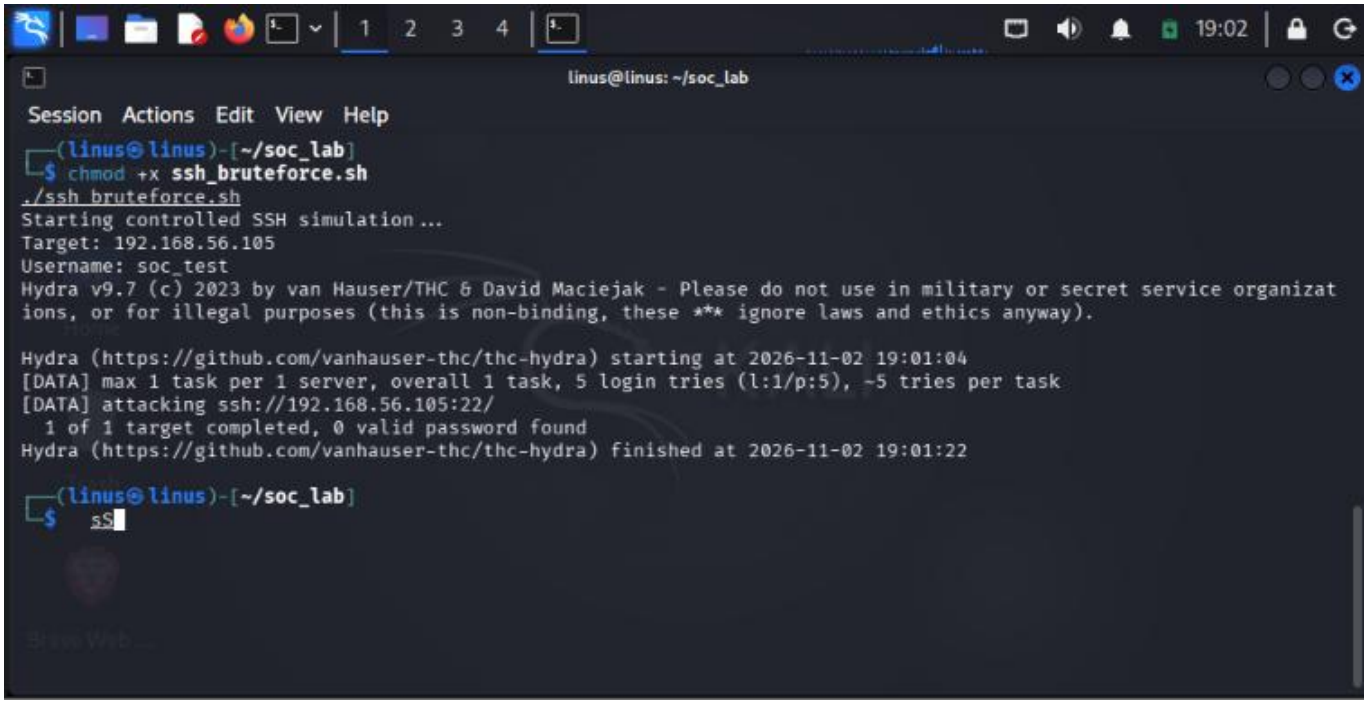


Figure: Execution of the controlled SSH brute-force simulation using Hydra from Kali Linux against Ubuntu02

4. Ubuntu02 Authentication Log Verification

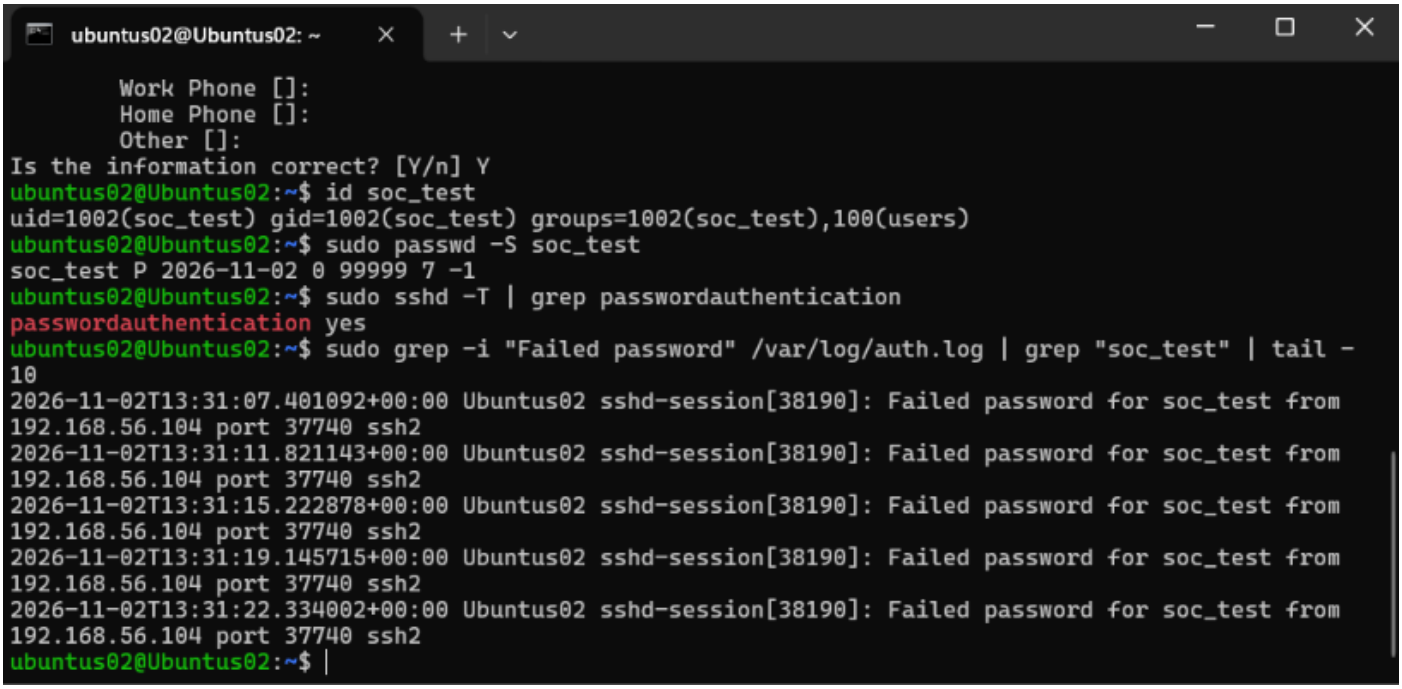
After executing the SSH brute-force simulation from Kali Linux, the Ubuntu02 terminal was used to inspect the system authentication log and verify that the generated failed SSH login attempts were successfully recorded.

The following command was used to retrieve the failed authentication events associated with the soc_test account:

sudo grep "Failed password" /var/log/auth.log | grep "soc_test" | tail -10

The terminal output displayed five failed SSH login attempts originating from the Kali Linux machine (192.168.56.104) and targeting the soc_test account on Ubuntu02.

This verification confirmed that the target system successfully recorded the authentication failures generated during the controlled simulation. These events were then available for forwarding to Splunk Enterprise for further security event analysis.



Verification of failed SSH authentication events in the Ubuntu02 authentication log

5. Analysis of Events in Splunk Enterprise

After verifying the failed SSH authentication events in the Ubuntu02 authentication log, the forwarded events were examined in Splunk Enterprise using the Search & Reporting interface.

The initial search was performed to verify successful log collection and identify the failed SSH login activities. The returned events were reviewed to examine timestamps, target host, source IP address and authentication details.

5.1 Initial Event Search

The following SPL query was used to retrieve the failed SSH authentication events:

```
index=linux_system "Failed password"
| table _time host source sourcetype _raw
| sort -_time
```

The search results displayed the collected authentication events, allowing the event details to be examined.

5.2 Event Extraction and Analysis

A further SPL search was performed to extract and summarize the failed login attempts by source IP address, targeted username and host.

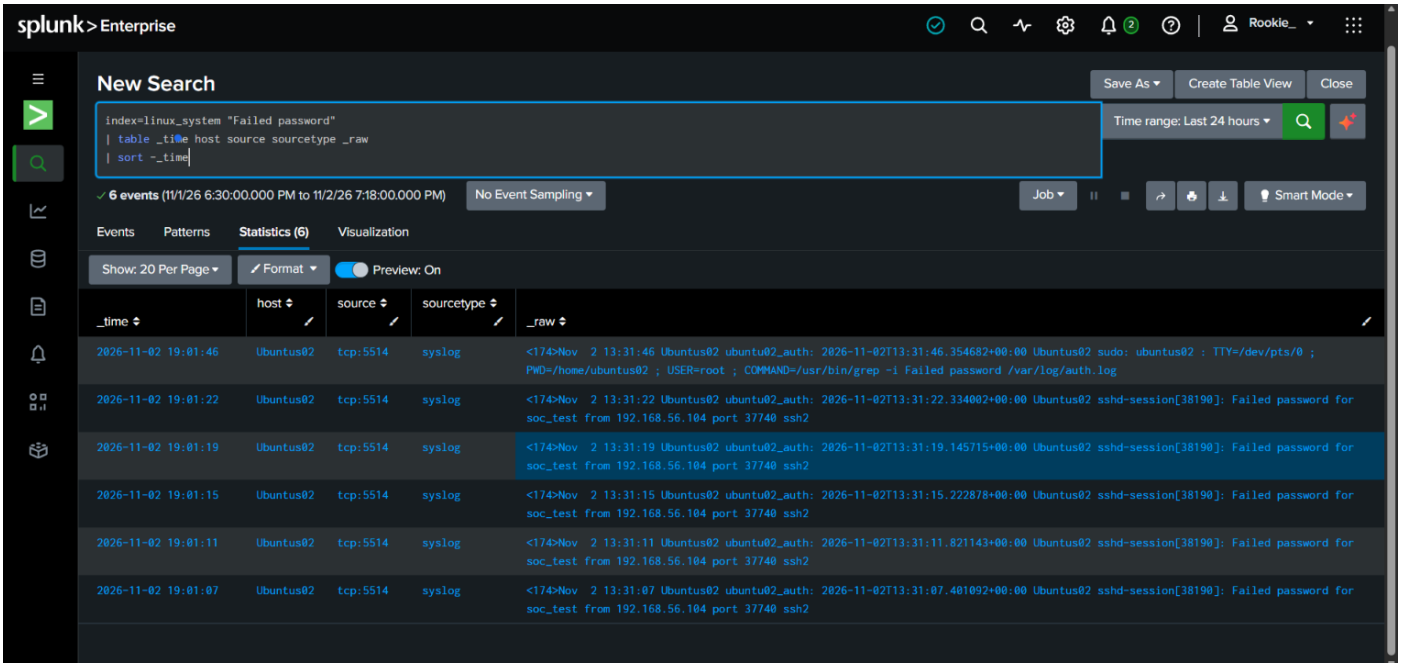
```
index=linux_system sourcetype=syslog "Failed password"
| rex field=_raw "Failed password for (?:invalid user )?(?<target_username>\S+) from (?<source_ip>\d+\.\d+\.\d+\.\d+)"
```

```
| stats count as failed_attempts
      min(_time) as first_attempt
      max(_time) as last_attempt
      by source_ip target_username host
| convert ctime(first_attempt) ctime(last_attempt)
```

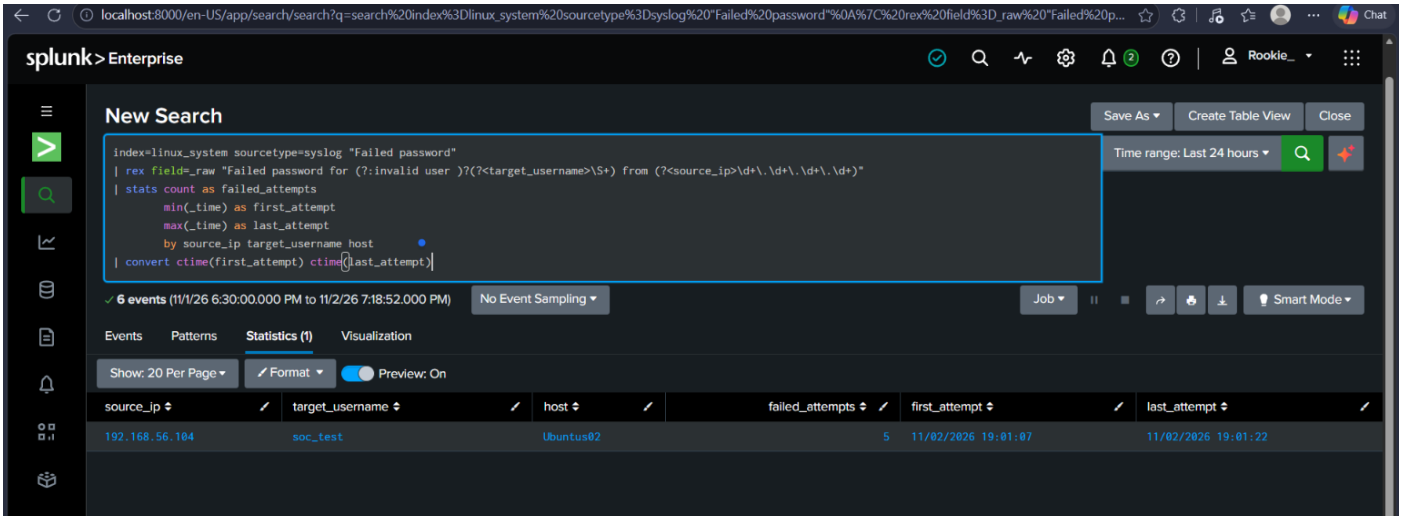
5.3 Analysis Results

The analysis identified five failed SSH login attempts originating from Kali Linux (192.168.56.104) and targeting the soc_test account on Ubuntu02 (Ubuntus02).

The results confirmed that the authentication events were successfully collected in Splunk Enterprise and that the source IP, targeted username, target host and event timestamps could be extracted and summarized using SPL.



Initial failed SSH authentication event search in Splunk Search & Reporting.



SPL field extraction and aggregation of failed SSH login attempts

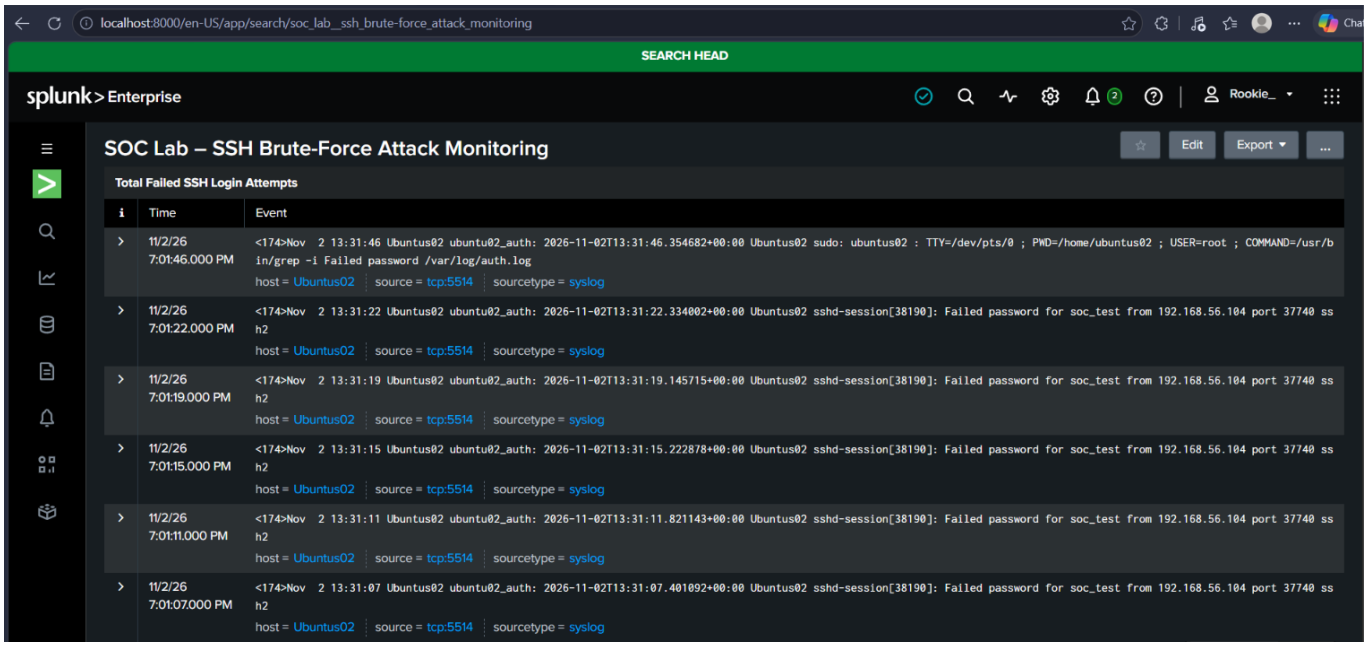
6. SOC Dashboard Development

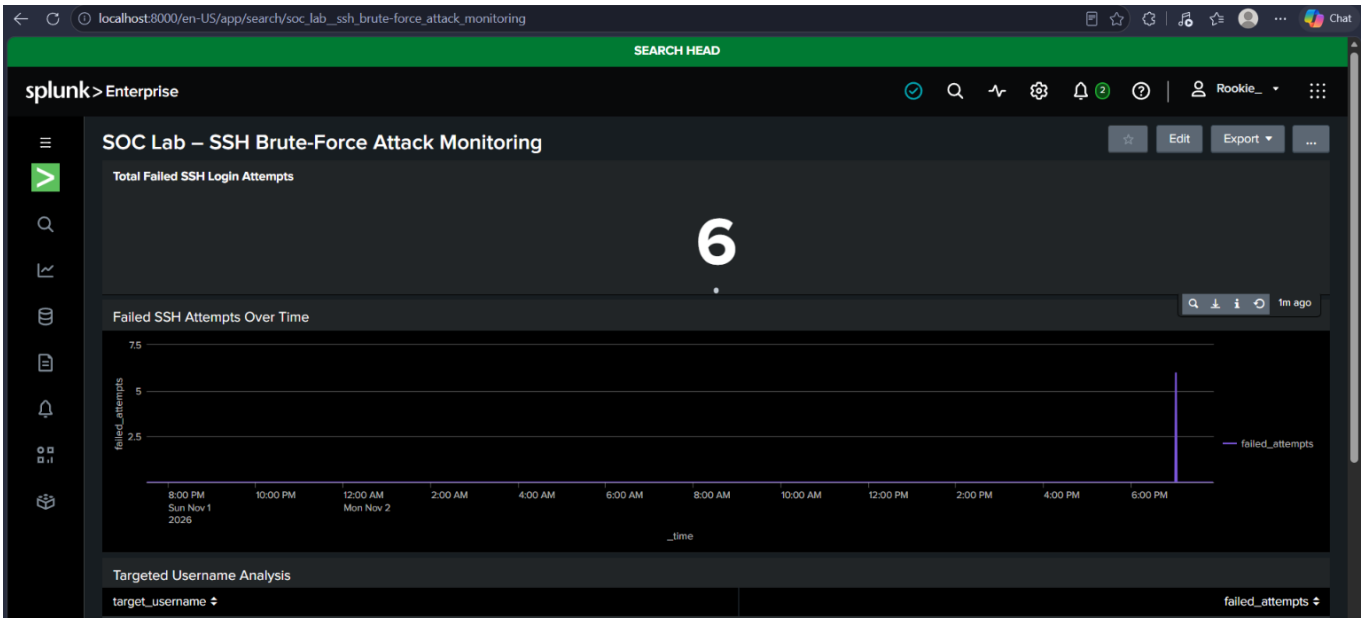
A dedicated Splunk dashboard named **SOC Lab – SSH Brute-Force Attack Monitoring** was created to visualize and monitor the SSH authentication activity detected in the lab environment.

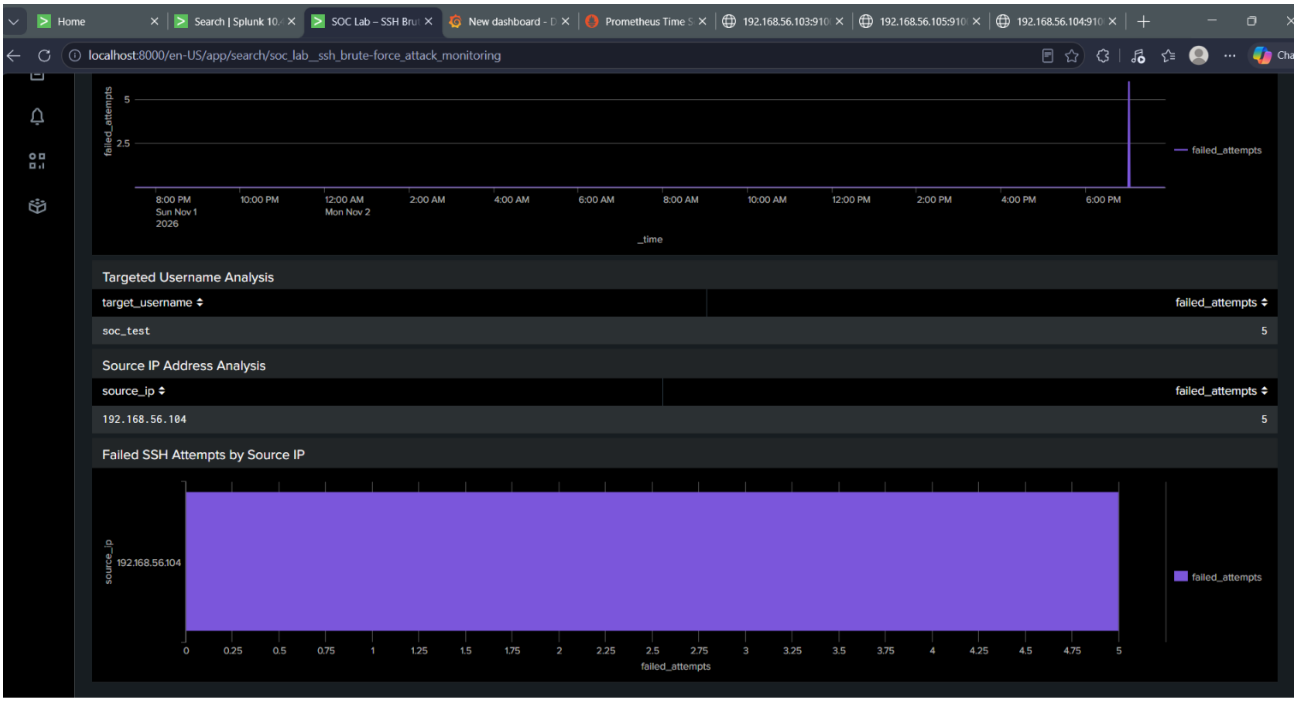
The dashboard contains the following panels:

- **Total Failed SSH Login Attempts:** Displays the total number of matching events.
- **Failed SSH Attempts Over Time:** Visualizes the occurrence of failed login attempts over time.
- **Targeted Username Analysis:** Identifies the username targeted during the SSH login attempts.
- **Source IP Address Analysis:** Displays the IP address from which the failed login attempts originated.
- **Failed SSH Attempts by Source IP:** Provides a graphical representation of failed attempts grouped by source IP address.

The dashboard provides a centralized view of the authentication activity, making it easier to identify the source of suspicious login attempts, the targeted account and the frequency of failed authentication events.







SOC Lab – SSH Brute-Force Attack Monitoring dashboard.

7. Scheduled Alert Configuration

A scheduled alert named SSH Brute-Force Attack Detected was created in Splunk Enterprise to identify repeated failed SSH login attempts.

The scheduled alert was configured to execute every minute using the Cron expression * * * * *. The search time range was set to Last 24 hours, while the alert trigger condition was configured to activate when the detection search returned at least one result. The five-minute throttling period was enabled to reduce repeated notifications.

with the following settings:

Schedule and Trigger Conditions

Setting	Configuration
Alert Type	Scheduled
Schedule	Cron Schedule
Cron Expression	* * * * * (every minute)
Search Time Range	Last 24 hours
Alert Expiration	5 minutes
Trigger Condition	Number of Results greater than 0
Trigger Frequency	Once
Throttling	Enabled
Suppression Period	300 seconds (5 minutes)

The alert was configured to trigger when the detection search returned at least one result matching its configured detection threshold of five or more failed SSH login attempts.

The trigger frequency was set to Once to avoid generating a separate notification for every individual result. Throttling was enabled with a five-minute suppression period to reduce repeated notifications.

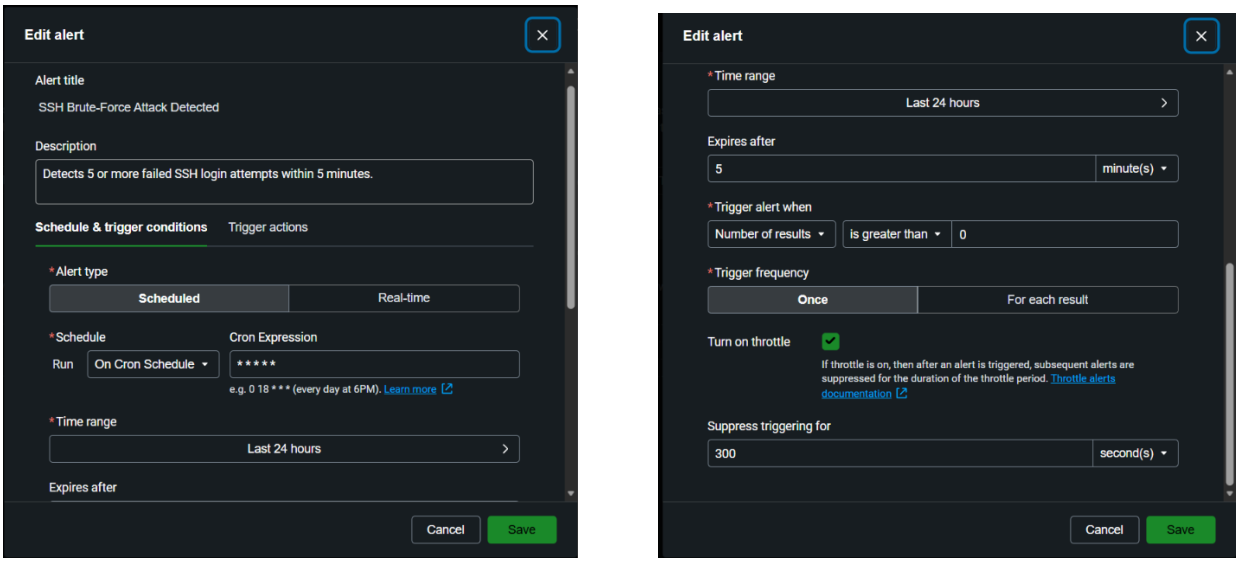


Figure: Splunk Enterprise – Scheduled alert configuration and trigger conditions.

7.1 Alert Execution and Email Notification Verification

After configuring the scheduled alert, the SSH brute-force simulation was executed again from Kali Linux using the custom Bash script and Hydra tool. The simulation generated another set of failed SSH login attempts against Ubuntu02.

The following activities were verified:

- **Attack Simulation: Hydra executed SSH login attempts against the Ubuntu02 target machine.**
- **Alert Status: The SSH Brute-Force Attack Detected alert was visible on the Splunk Alerts page with its status set to Enabled.**
- **Dashboard Monitoring: The SOC dashboard reflected the newly generated authentication activity, showing an increase in failed login events.**
- **Email Notification: Splunk successfully executed the configured email action and delivered a security notification to the configured Gmail account.**

The received email contained the detection name, severity, source IP address, targeted username, target host, failed login attempt count, first and last attempt timestamps, and detection threshold.
This verified the alert execution and email notification workflow within the SOC lab environment.

The dashboard, email notification and detailed investigation searches were executed at different stages of the simulation and may display different event counts due to their respective search conditions and execution periods. The detailed investigation identified 25 failed SSH password events associated with the targeted soc_test account.

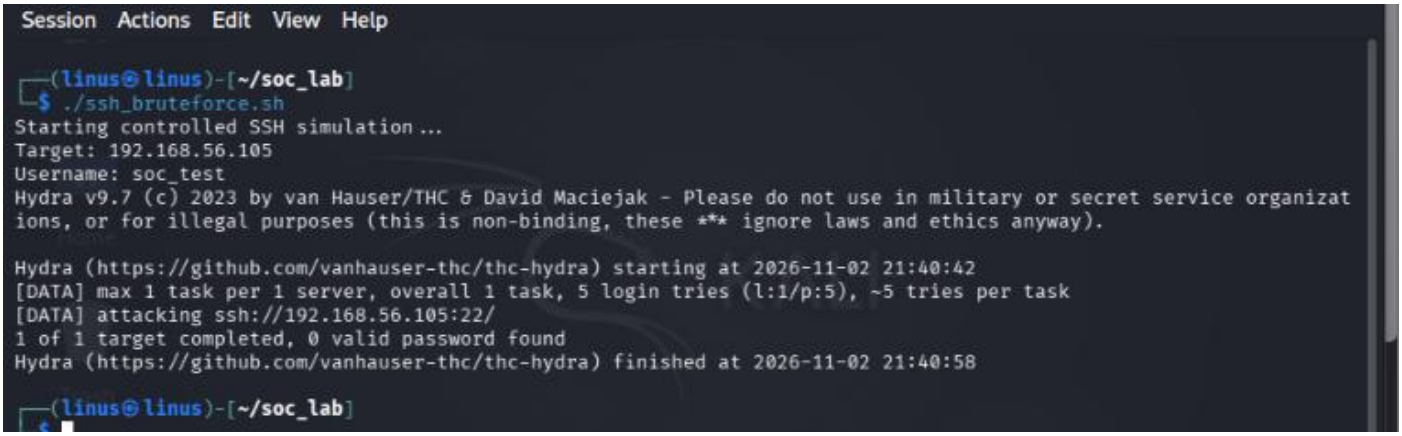


Figure: Hydra executed SSH login attempts against the Ubuntu02 target machine

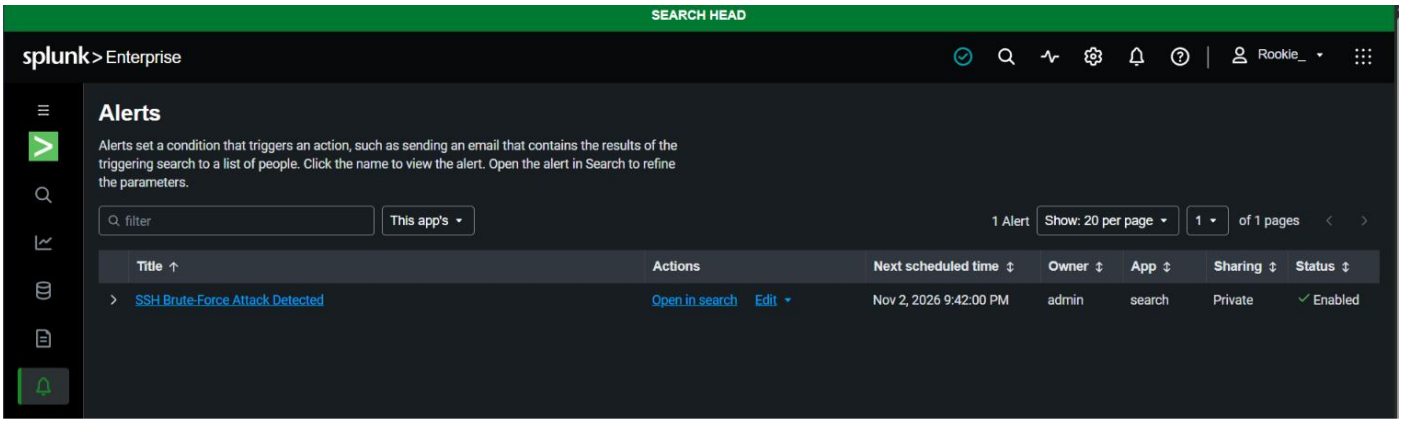


Figure: The SSH Brute-Force Attack Detected alert was visible on the Splunk Alerts page with its status set to Enabled

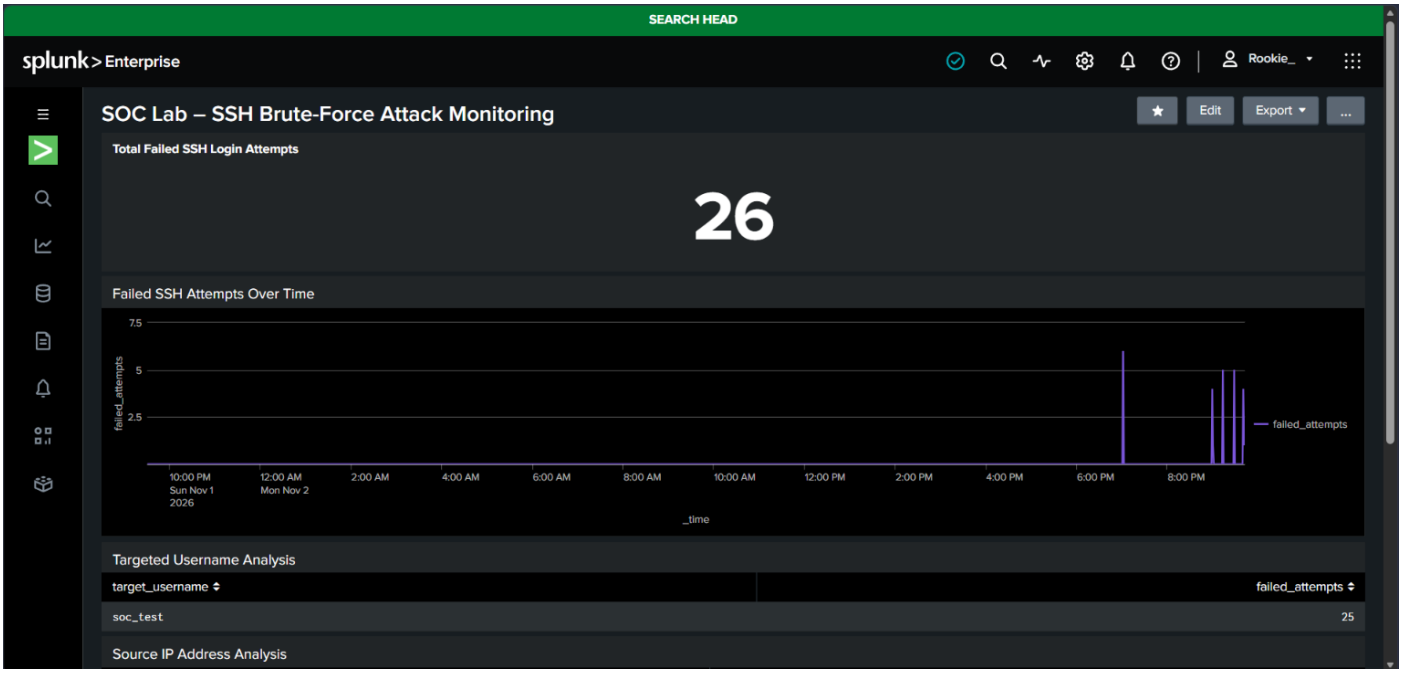


Figure: The SOC dashboard reflected the newly generated authentication activity, showing an increase in failed login events

SECURITY INCIDENT ALERT

A potential SSH brute-force attack has been detected by the SOC monitoring system.

Detection Name: SSH Brute-Force Attack Detected
Severity: Medium

ATTACK DETAILS
Source IP Address: 192.168.56.104
Target Username: soc_test
Target Host: Ubuntus02
Failed Login Attempts: 20

TIME DETAILS
First Attempt: 11/02/2026 19:01:07
Last Attempt: 11/02/2026 21:28:46

Detection Threshold: 5 or more failed attempts within 5 minutes.

Figure: Hydra SSH simulation, enabled Splunk alert, updated SOC dashboard and successfully received email notification.

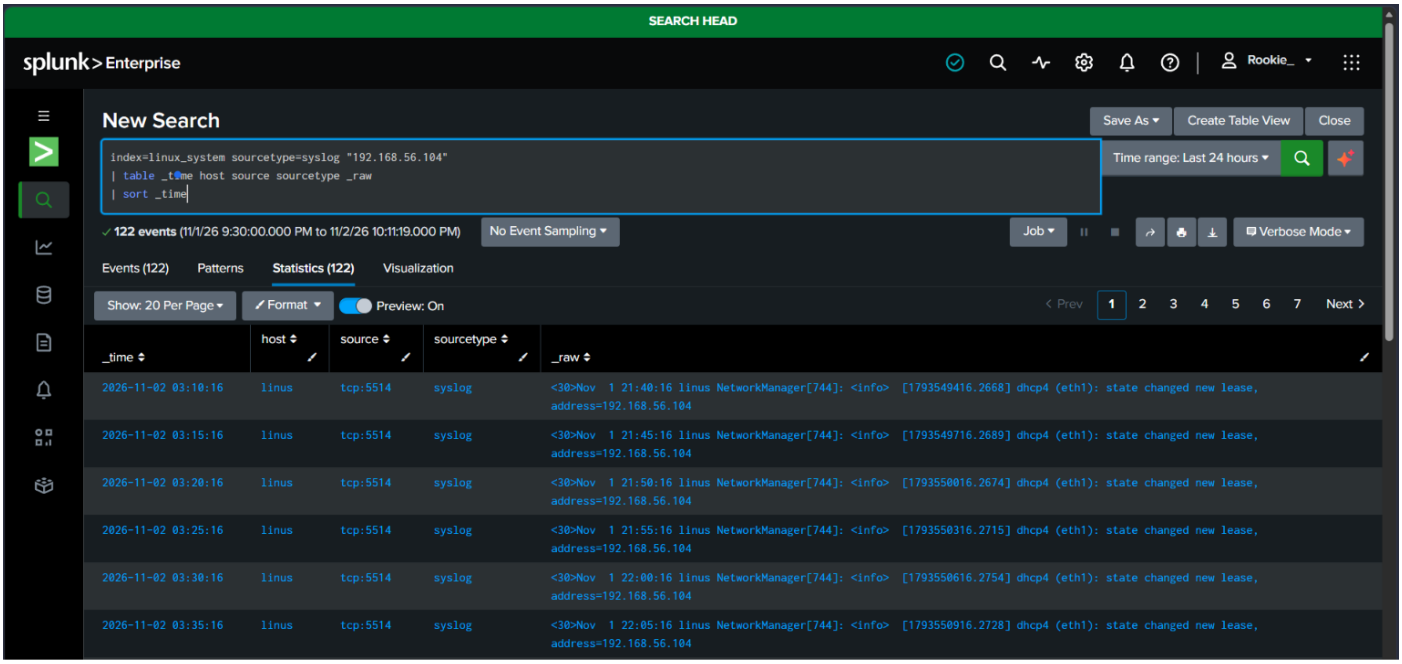
8. Incident Investigation

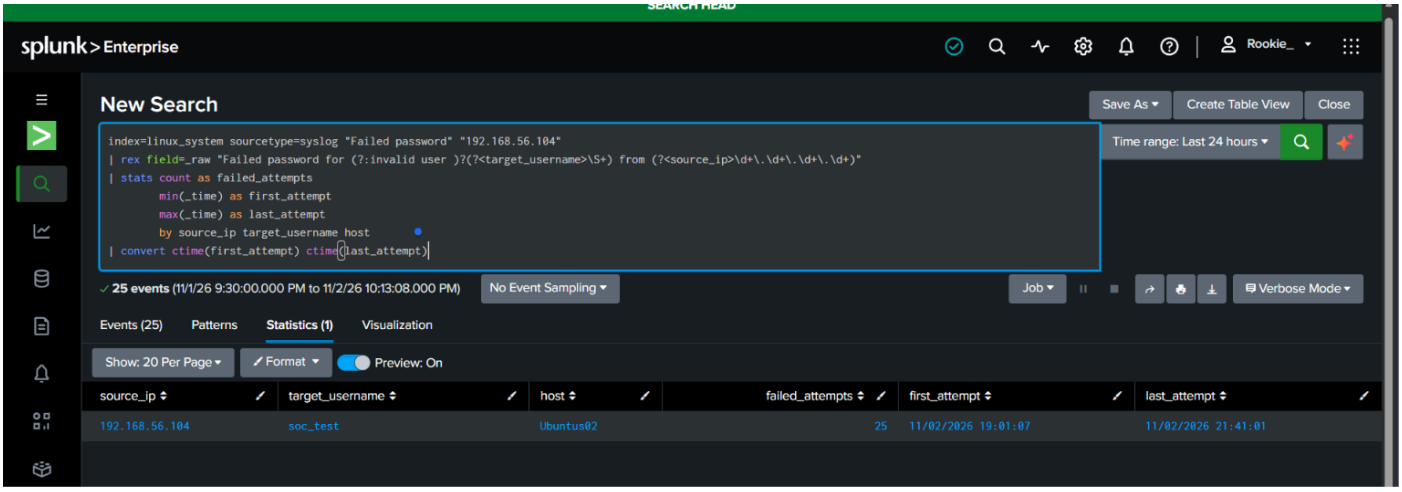
Following the alert notification, the collected authentication events were investigated in Splunk Enterprise to understand the source and nature of the suspicious activity.

8.1 Source IP and Authentication Activity Analysis

The authentication events were examined to identify the source IP address, targeted account and frequency of failed login attempts.

The analysis identified repeated failed SSH authentication attempts originating from the Kali Linux machine and targeting the soc_test account on Ubuntu02.

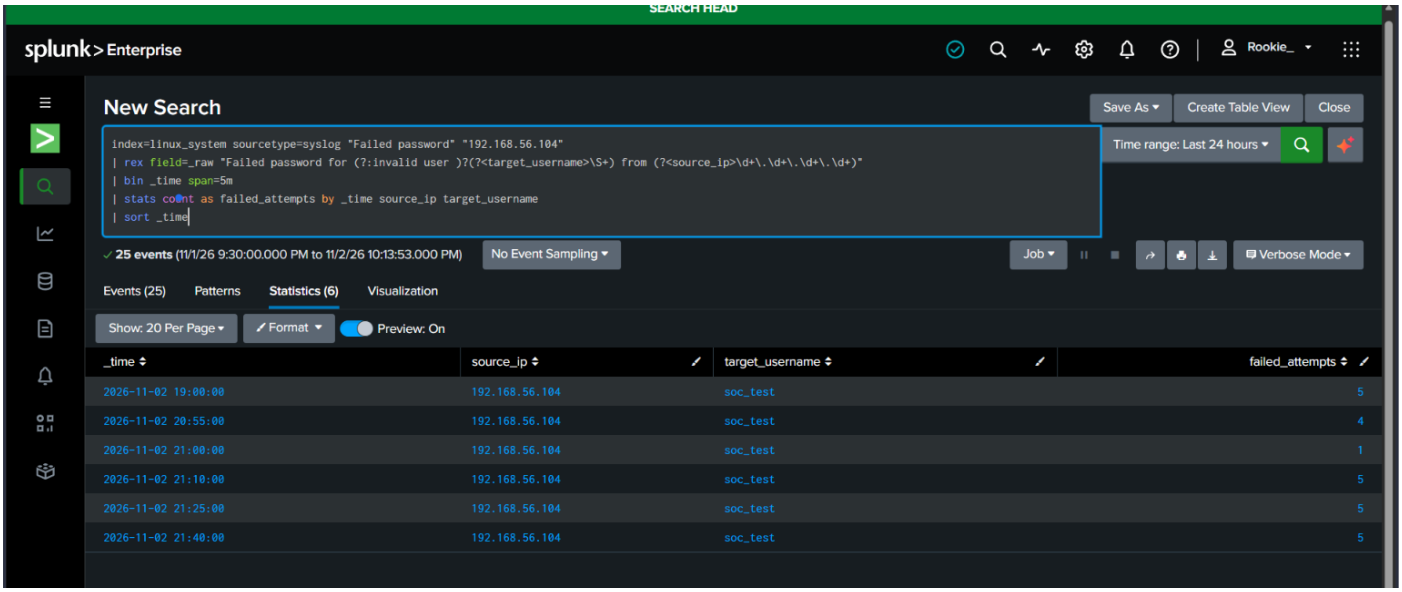




8.2 Failed Authentication Activity Analysis

The investigation identified a total of 25 failed SSH password events during the later simulation and investigation period. The events were associated with source IP address 192.168.56.104 and the targeted account soc_test.

The five-minute interval analysis showed repeated bursts of failed authentication activity during the investigation period.

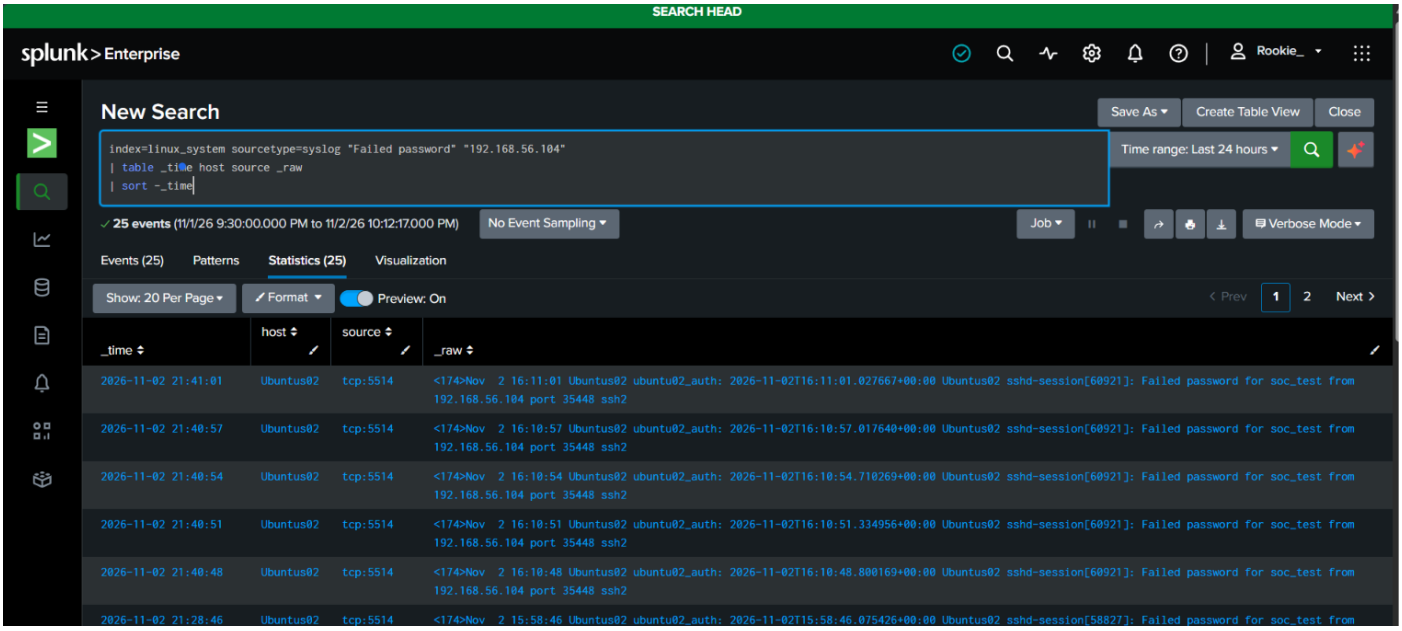


8.3 Successful Authentication Verification

A separate Splunk search was performed to check for successful SSH authentication events associated with the targeted soc_test account.

The search returned zero matching events within the selected 24-hour time range. No successful SSH login event for the targeted account was identified in the collected logs during this investigation period.

This verification helped distinguish the repeated failed authentication activity from any confirmed successful login activity.



8.4 Account Activity Analysis

A further investigation was performed to classify the authentication-related events associated with the targeted soc_test account.

The SPL query categorized the returned events into failed password attempts, password verification failures, connection closures, PAM failure summaries and other account-related activity.

The analysis returned 79 matching events, including:

- **Failed Password: 25 events**
- **Password Check Failed: 25 events**
- **Connection Closed: 5 events**
- **PAM Failure Summary: 5 events**
- **Other Account Activity: 19 events**

This classification provided additional visibility into how the Ubuntu02 system recorded and handled the repeated SSH authentication failures. These categories represent authentication-related log events and should not be interpreted as 79 separate login attempts.

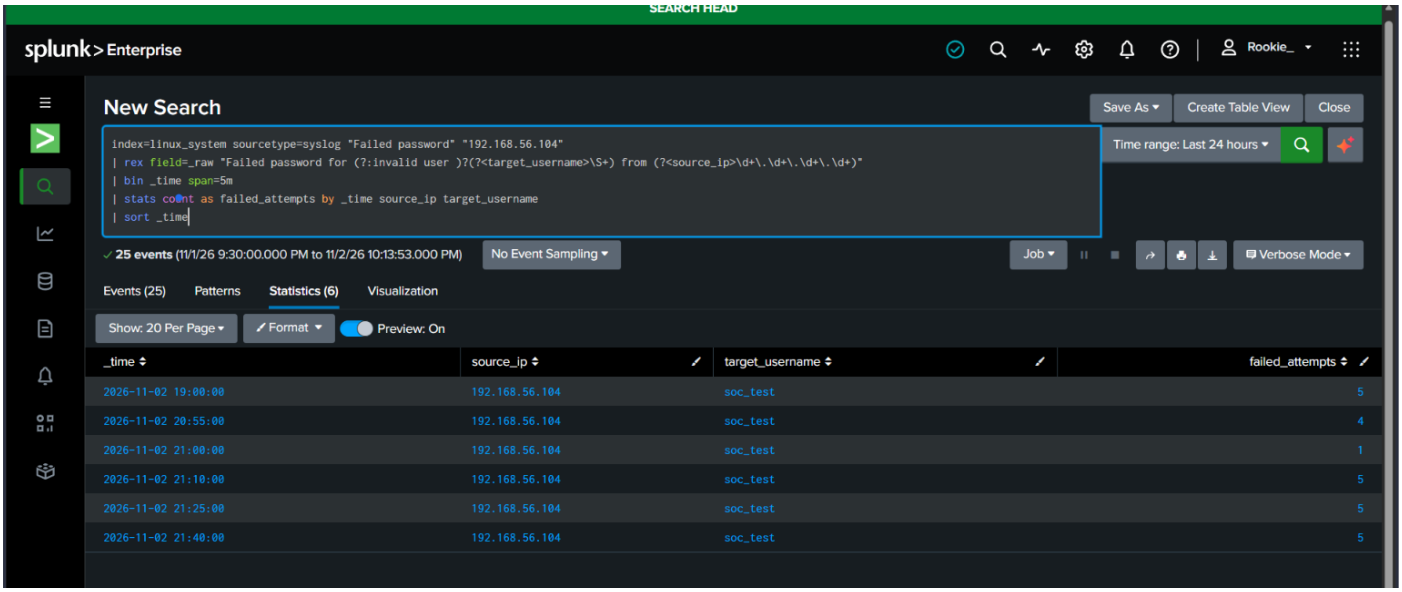


Figure 8.1: Classification and count of authentication-related events associated with the soc_test account in Splunk Enterprise.

8.5 Final Investigation Findings and Verdict

Based on the investigation conducted in Splunk Enterprise, repeated SSH authentication failures were identified on the Ubuntu02 target system.

The investigation established the following findings:

Investigation Detail	Finding
Source IP Address	192.168.56.104 (Kali Linux)
Target Host	Ubuntu02 (Ubuntus02)
Target Account	soc_test
Total Failed SSH Attempts	25
Successful Authentication Events	No matching events identified within the selected 24-hour investigation period
Authentication Activity	Failed password attempts, password verification failures, connection closures and PAM failure summaries
Detection Status	Splunk alert successfully triggered and an email notification was received

Final Verdict: Confirmed SSH Brute-Force Simulation – No Successful Compromise Observed.

The investigation confirmed that the activity originated from the Kali Linux machine used for the controlled attack simulation within the isolated SOC lab environment. The Ubuntu02 authentication logs recorded the failed login attempts, which were successfully forwarded to Splunk Enterprise for monitoring, analysis and detection.

No matching successful SSH authentication event was identified for the targeted account during the selected investigation period. Therefore, based on the available evidence, there was no indication of a successful account compromise in this simulation.

The detection, alerting and email notification workflow was successfully demonstrated as part of the SOC lab project.

9. Incident Response Documentation

9.1 Incident Summary

A controlled SSH brute-force simulation was performed against the Ubuntu02 target system using Kali Linux. Splunk Enterprise detected repeated failed SSH authentication attempts and generated an alert with an email notification.

Incident Detail	Information
Incident Name	SSH Brute-Force Activity
Source IP	192.168.56.104
Source System	Kali Linux

Target System	**Ubuntu02**
Target Account	**soc_test**
Total Failed SSH Attempts	**25**
Detection Platform	**Splunk Enterprise**
Alert Severity	**Medium**
Incident Type	**SSH Brute-Force Simulation**

9.2 Evidence Collected

The following evidence was collected during the investigation:

1. **Kali Linux terminal output showing the Hydra-based SSH simulation.**
2. **Ubuntu02 authentication logs showing failed SSH login attempts.**
3. **Splunk search results containing the forwarded authentication events.**
4. **SPL analysis identifying the source IP, targeted username and failed attempt count.**
5. **Five-minute interval analysis showing repeated bursts of failed authentication.**
6. **SOC dashboard displaying the authentication activity.**
7. **Splunk alert configuration and enabled alert status.**
8. **Email notification containing the incident details.**

9.3 Recommended Incident Response Actions

Based on the investigation findings, the following response actions are recommended for a similar incident in a production environment:

- **Validate the source IP address and identify the originating system.**
- **Review authentication logs for repeated failed login attempts.**
- **Verify whether any successful login occurred during the investigation period.**
- **Review the targeted account for suspicious activity.**
- **Consider temporarily restricting the source IP if malicious activity is confirmed.**
- **Strengthen SSH authentication using key-based authentication and appropriate access controls.**
- **Continue monitoring authentication events through the SIEM platform.**

9.4 Incident Outcome

The controlled SSH brute-force simulation was successfully detected by Splunk Enterprise. The generated authentication events were collected, analyzed and visualized through the SOC dashboard.

The configured scheduled alert triggered, and the email notification was successfully delivered to the configured Gmail account.

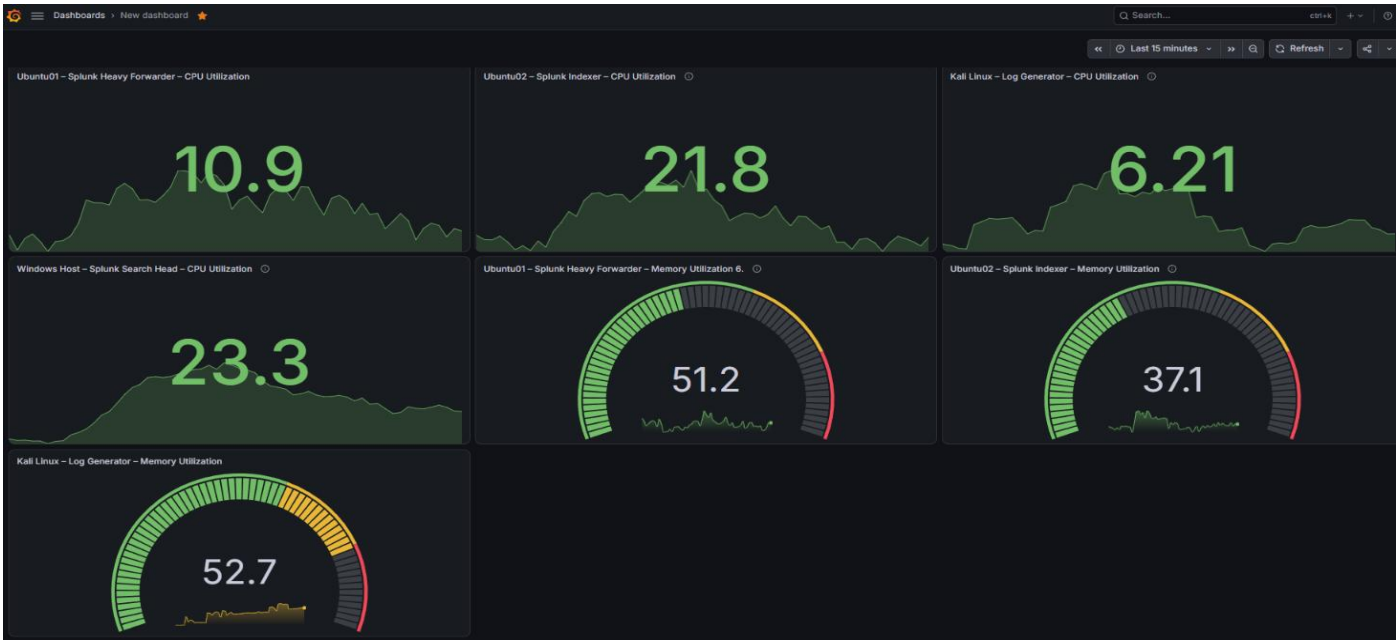
No matching successful SSH authentication event was identified for the targeted account within the selected 24-hour investigation period.

Final Incident Status: Controlled SSH Brute-Force Simulation Detected – No Successful Compromise Observed.

10. Infrastructure Monitoring – Prometheus and Grafana

Prometheus and Grafana were integrated into the SOC lab environment to monitor the performance and health of the virtual machines. The monitoring setup provided visibility into CPU utilization and memory usage across the monitored systems through Grafana dashboards, using Prometheus as the metrics collection and querying platform.

The screenshots below demonstrate the collected infrastructure metrics and their visualization through Grafana, providing an overview of the resource utilization of the lab environment.



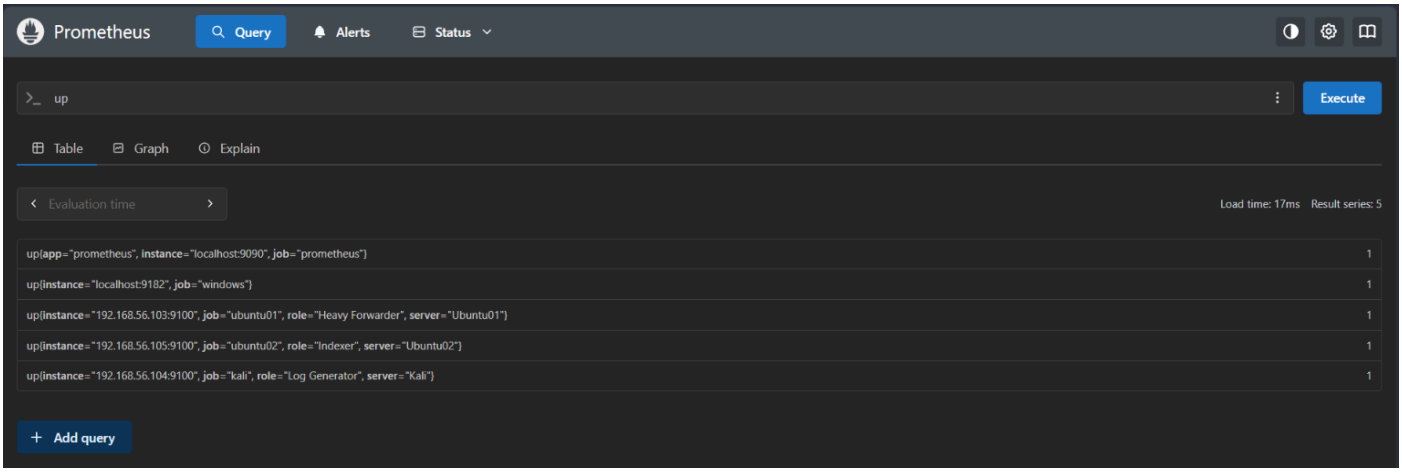


Figure: Prometheus and Grafana infrastructure monitoring setup.

11. Project Conclusion

This project demonstrated an end-to-end SOC monitoring and detection workflow using a distributed Splunk Enterprise environment.

The implementation covered log collection, centralized monitoring, SSH attack simulation, security event analysis, SPL-based detection, dashboard development, scheduled alerting, email notification, incident investigation and infrastructure monitoring.

The project provided practical experience in monitoring authentication events, identifying suspicious activity, analyzing security logs and documenting investigation findings using SIEM capabilities.